# ROMÁN DEL ANGEL JUÁREZ

**Associate AI Engineer • Conversational AI & Enterprise Integrations**

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**LinkedIn Profile**

## PROFESSIONAL SUMMARY

AI Engineer in early career with hands-on experience building **AI-powered assistants, LLM integrations, and knowledge-grounded question-answering solutions** that connect AI capabilities with existing enterprise platforms. Currently part of an AI engineering team delivering language model solutions that improve how users access institutional information and streamline workflows. Experienced in **integrating the OpenAI API, fine-tuning MiniLM for intent classification and entity extraction, preparing documents for retrieval, and building backend services** with Python, JavaScript, Node.js, and SQL. Careful about **answer quality and responsible AI**: generated responses are validated against approved documentation.

## PROFESSIONAL EXPERIENCE

### Junior AI Developer — ClusterIA Department

*Oct 2025 – Present*

*Instituto de Estudios Superiores de Tamaulipas*

- Develop **AI-powered applications** and knowledge bases that support institutional workflows, natural-language queries, and information retrieval.
- Develop and maintain a **Canvas LMS LTI integration** from scratch using JavaScript and the **OpenAI API**, connecting AI capabilities with an existing enterprise platform used by end users.
- Fine-tune a **MiniLM language model** for natural language understanding, including intent classification and entity extraction, so the assistant interprets user requests more accurately.
- Prepare and transform source content for AI use through **document ingestion, vector stores, and retrieval-based question answering** experiments.
- Test AI solutions by **validating generated answers against approved documentation**, and investigate and resolve issues when they arise, supporting quality and responsible use of AI.
- Contribute to the implementation and continuous refinement of AI solutions as part of a team.

### Freelance Backend Developer

*Jun 2025 – Oct 2025*

*ERM Project — College of Anesthesiology of Tampico, Tamaulipas*

- Designed and developed a custom **enterprise management system** to centralize organizational workflows and internal business processes.
- Implemented backend architecture, **database models, business logic, and system integrations** to support application requirements.
- Worked directly with stakeholders to gather requirements and managed the full development lifecycle, from **system design through implementation, testing, and delivery**.

## TECHNICAL SKILLS

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|---|---|
| **AI / ML:** | LLM integration, OpenAI API, conversational AI, MiniLM fine-tuning, NLP, intent classification, entity extraction |
| **RAG / Retrieval:** | Knowledge bases, document ingestion and processing, embeddings, vector stores, semantic retrieval, grounded question answering |
| **Integration / APIs:** | REST API integration, LTI integrations, enterprise system integration, backend development, business logic |
| **Programming:** | Python, JavaScript, Node.js, SQL, C++, PHP |
| **Data:** | SQL, MySQL, PostgreSQL, MongoDB, data preparation and transformation |

## EDUCATION

### B.S. in Computer Systems and Digital Business

*Expected Dec 2026*

*Instituto de Estudios Superiores de Tamaulipas*

*Ninth semester*

## LANGUAGES

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|---|---|
| **Spanish:** | Native |
| **English:** | Intermediate |